

딥러닝 기반 추천 시스템

모두의연구소

다양한 딥러닝 기반 추천 시스템: 특징과 목표

이수진

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01

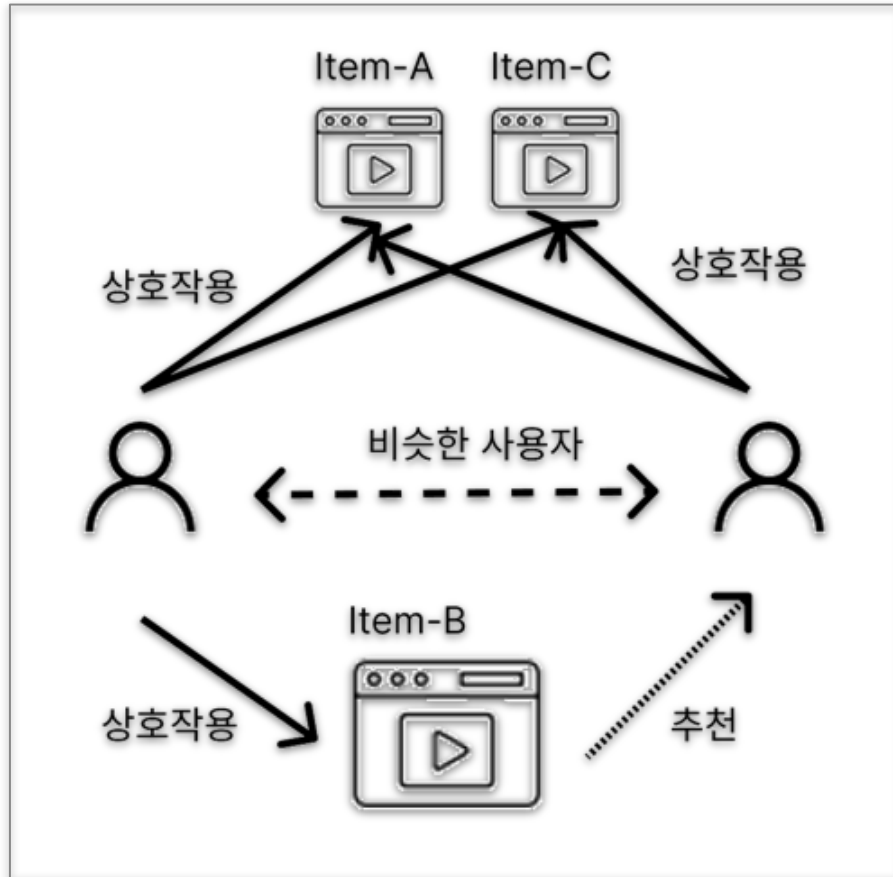
협업 필터링의 발전

1. CF와 NCF

1. CF와 NCF - CF

협업 필터링 기억하시나요?
Collaborative Filtering 말이죠!

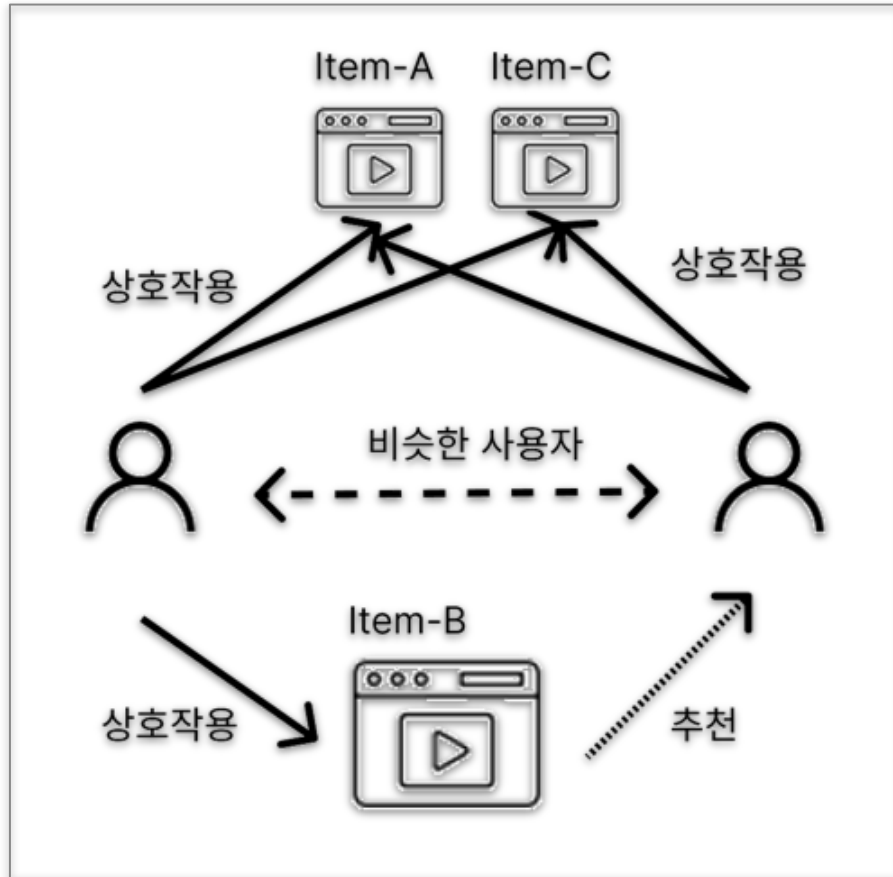
1. CF와 NCF - CF



CF로도 많이 불리는 협업 필터링 추천 시스템 방법,
정말 많이 사용되는 기술입니다.

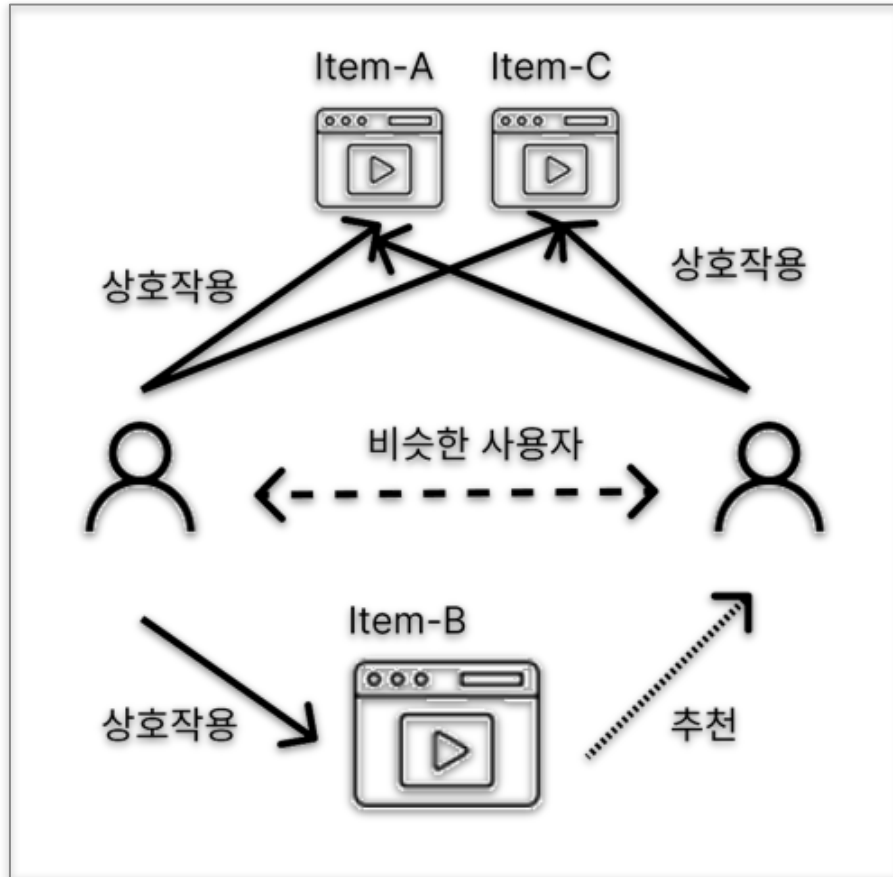
- 간단하지만, 강력한 성능을 제공해주죠
- 사용자와 아이템간의 상호작용을 분석해 다수의 사용자들의 패턴을 반영한 방법입니다.
- 비슷한 취향을 가진 사람끼리는 비슷한 콘텐츠를 소비할 것이다라는 가정을 가지고 있죠

1. CF와 NCF - CF



그러나,
한계점이 존재합니다.
무엇일까요?

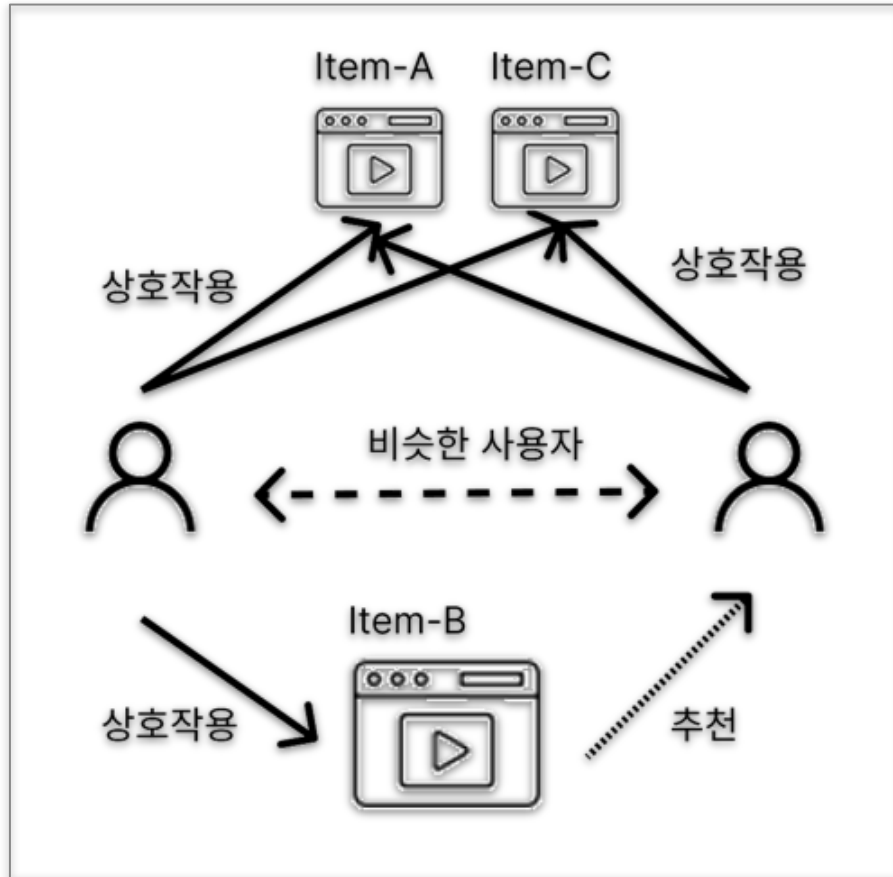
1. CF와 NCF - CF



- 아이템과 사용자 수가 많아지면 데이터가 굉장히 희소(sparse)해진다.
- 아이템 수와 사용자 수가 많아지면 연산량이 증가한다.
- 콜드 스타트(Cold start) 문제가 존재한다.
- 유명한 것들로 추천될 수 있다(Bias가 존재한다).

여러 한계점이 지적됩니다.

1. CF와 NCF



여기, 하나 흥미로운 논문이 있습니다.

1. CF와 NCF - NCF

Neural Collaborative Filtering*

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추천 시스템 분야에서
굉장히 유명한 논문이죠.

출처: Xiangnan He, [arxiv_NCF\(2017\)](#)

1. CF와 NCF - NCF

ABSTRACT

In recent years, deep neural networks have yielded immense success on speech recognition, computer vision and natural language processing. However, the exploration of deep neural networks on recommender systems has received relatively less scrutiny. In this work, we strive to develop techniques based on neural networks to tackle the key problem in recommendation — collaborative filtering — on the basis of implicit feedback.

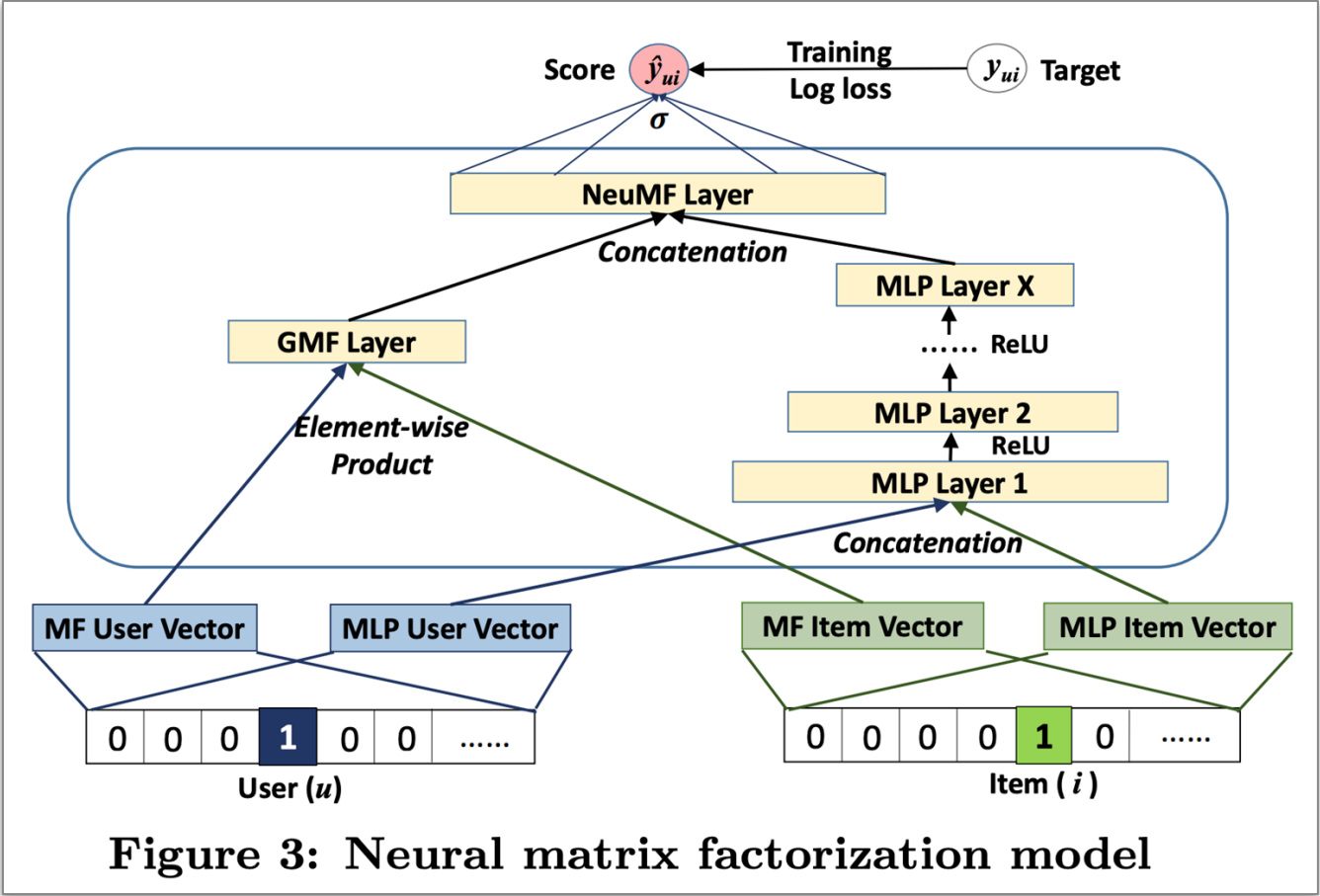
Although some recent work has employed deep learning for recommendation, they primarily used it to model auxiliary information, such as textual descriptions of items and acoustic features of musics. When it comes to model the key factor in collaborative filtering — the interaction between user and item features, they still resorted to matrix factorization and applied an inner product on the latent features of users and items.

By replacing the inner product with a neural architecture that can learn an arbitrary function from data, we present a general framework named NCF, short for *Neural network-based Collaborative Filtering*. NCF is generic and can express and generalize matrix factorization under its framework. To supercharge NCF modelling with non-linearities, we propose to leverage a multi-layer perceptron to learn the user-item interaction function. Extensive experiments on two real-world datasets show significant improvements of our proposed NCF framework over the state-of-the-art methods. Empirical evidence shows that using deeper layers of neural networks offers better recommendation performance.

- CF에 뉴럴넷을 적용하고 이때 암묵적 피드백(Implicit feedback) 활용
- 행렬 분해(Matrix Factorization)는 NCF의 스페셜 케이스
- 사용자-아이템 상호작용을 학습하기 위해 Multi Layer Perceptron 도입

출처: X. He, [arxiv_NCF\(2017\)](#)

1. CF와 NCF - NCF



출처: X. He, arxiv_NCF(2017)

1. CF와 NCF - NCF

Neural Collaborative Filtering*

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출처: X. He, [arxiv_NCF\(2017\)](#)

기존 CF의 한계점을 극복하기 위해
딥러닝을 이용한 구조를 제안합니다.

실제로 성능도 정말 좋습니다

1. CF와 NCF - NCF

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[출처: X. He, arxiv_NCF\(2017\)](#)

**협업 필터링과 행렬 분해 방법은
현재까지도 계속 발전하고 있고
다양하게 응용되고 있습니다!**

1. CF와 NCF - NCF

Neural Collaborative Filtering*

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출처: X. He, [arxiv_NCF\(2017\)](#)

이 논문의 자세한 리뷰는
4장에서 살펴보겠습니다!

Shallow 영역과 Deep 영역

1. Shallow와 Deep의 융합

1. Shallow와 Deep의 융합

Shallow와 Deep?
이게 무슨 말일까요?

1. Shallow와 Deep의 융합

- Shallow(얕은)
 - 선형적 방법, 얕은 모델, 저차 상호작용(**Low Order Interaction**, 1~2 order)
 - (남자-20대), (여자-20대), (20대-서울) ...
 - Factorization Machine(FM), Logistic Regression(LR)
- Deep(깊은)
 - Deep 구조를 가진 방법, 고차 상호작용(**High Order Interaction**, 3~N order)
 - (남자-20대-한국), (여자-20대-뉴욕), (여자-30대-한국-서울) ...
 - DNN, Attention 등

1. Shallow와 Deep의 융합 - FM

Factorization Machines

Steffen Rendle

Department of Reasoning for Intelligence
The Institute of Scientific and Industrial Research
Osaka University, Japan
rendle@ar.sanken.osaka-u.ac.jp

출처 : [S. Rendle, IEEE-FM\(2010\)](#)

추천 시스템 분야에서
굉장히 유명한 논문입니다.
NCF만큼 유명합니다.
Shallow(Low Order) 방법의 대표이죠.

1. Shallow와 Deep의 융합 - FM

Feature vector $\mathbf{x}$																	Target y					
$\mathbf{x}^{(1)}$	1	0	0	...	1	0	0	0	...	0.3	0.3	0.3	0	...	13	0	0	0	0	...	5	$y^{(1)}$
$\mathbf{x}^{(2)}$	1	0	0	...	0	1	0	0	...	0.3	0.3	0.3	0	...	14	1	0	0	0	...	3	$y^{(2)}$
$\mathbf{x}^{(3)}$	1	0	0	...	0	0	1	0	...	0.3	0.3	0.3	0	...	16	0	1	0	0	...	1	$y^{(2)}$
$\mathbf{x}^{(4)}$	0	1	0	...	0	0	1	0	...	0	0	0.5	0.5	...	5	0	0	0	0	...	4	$y^{(3)}$
$\mathbf{x}^{(5)}$	0	1	0	...	0	0	0	1	...	0	0	0.5	0.5	...	8	0	0	1	0	...	5	$y^{(4)}$
$\mathbf{x}^{(6)}$	0	0	1	...	1	0	0	0	...	0.5	0	0.5	0	...	9	0	0	0	0	...	1	$y^{(5)}$
$\mathbf{x}^{(7)}$	0	0	1	...	0	0	1	0	...	0.5	0	0.5	0	...	12	1	0	0	0	...	5	$y^{(6)}$
	A	B	C	...	TI	NH	SW	ST	...	TI	NH	SW	ST	...	Time	TI	NH	SW	ST	...		
	User				Movie					Other Movies rated						Last Movie rated						

Fig. 1. Example for sparse real valued feature vectors $\mathbf{x}$ that are created from the transactions of example 1. Every row represents a feature vector $\mathbf{x}^{(i)}$ with its corresponding target $y^{(i)}$. The first 4 columns (blue) represent indicator variables for the active user; the next 5 (red) indicator variables for the active item. The next 5 columns (yellow) hold additional implicit indicators (i.e. other movies the user has rated). One feature (green) represents the time in months. The last 5 columns (brown) have indicators for the last movie the user has rated before the active one. The rightmost column is the target – here the rating.

출처 : S. Rendle, IEEE-FM(2010)

- input 데이터는 feature vector $\mathbf{x}$ 형식으로 고려
 - feature vector $\mathbf{x}$ 는 희소(sparse)함
 - 이미지는 그 예시를 보여주는 것
- 파란색: 사용자를 의미하는 벡터
- 주황색: 아이템을 나타내는 벡터
- 노란색: 사용자가 부여한 다른 영화 평점
- 초록색: 월 단위 시간 값
- 갈색: 사용자가 해당 영화를 평가하기 직전에 평가한 영화

1. Shallow와 Deep의 융합 - FM

A. Factorization Machine Model

1) *Model Equation*: The model equation for a factorization machine of degree $d = 2$ is defined as:

$$\hat{y}(\mathbf{x}) := w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=i+1}^n \langle \mathbf{v}_i, \mathbf{v}_j \rangle x_i x_j \quad (1)$$

where the model parameters that have to be estimated are:

$$w_0 \in \mathbb{R}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \mathbf{V} \in \mathbb{R}^{n \times k} \quad (2)$$

And $\langle \cdot, \cdot \rangle$ is the dot product of two vectors of size k :

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle := \sum_{f=1}^k v_{i,f} \cdot v_{j,f} \quad (3)$$

A row $\mathbf{v}_i$ within $\mathbf{V}$ describes the i -th variable with k factors. $k \in \mathbb{N}_0^+$ is a hyperparameter that defines the dimensionality of the factorization.

A 2-way FM (degree $d = 2$) captures all single and pairwise interactions between variables:

- w_0 is the global bias.
- w_i models the strength of the i -th variable.
- $\hat{w}_{i,j} := \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ models the interaction between the i -th and j -th variable. Instead of using an own model parameter $w_{i,j} \in \mathbb{R}$ for each interaction, the FM models the interaction by factorizing it. We will see later on, that this is the key point which allows high quality parameter estimates of higher-order interactions ($d \geq 2$) under sparsity.

출처 : S. Rendle, IEEE-FM(2010)

1. Shallow와 Deep의 융합 - FM

A. Factorization Machine Model

1) *Model Equation*: The model equation for a factorization machine of degree $d = 2$ is defined as:

$$\hat{y}(\mathbf{x}) := w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=i+1}^n \langle \mathbf{v}_i, \mathbf{v}_j \rangle x_i x_j \quad (1)$$

where the model parameters that have to be estimated are:

$$w_0 \in \mathbb{R}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \mathbf{V} \in \mathbb{R}^{n \times k} \quad (2)$$

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A 2-way FM (degree $d = 2$) captures all single and pairwise interactions between variables:

1. feature마다 가중치를 부여한다

2. feature들 간의 상호작용을 고려한

a. 이때 내적을 활용한다
(dot product)

출처 : S. Rendle, IEEE-FM(2010)

1. Shallow와 Deep의 융합 - FM

A. Factorization Machine Model

1) **사용자 성별, 장르 등 다양한 변수를**
 1) Model Equation: The model equation for a factorization machine of degree $d = 2$ is defined as:

$$f(x) = w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=i+1}^n \langle \mathbf{v}_i, \mathbf{v}_j \rangle x_i x_j \quad (1)$$

where the model parameters that have to be estimated are:

$$w_0 \in \mathbb{R}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \mathbf{V} \in \mathbb{R}^{n \times k} \quad (2)$$

And $\langle \cdot, \cdot \rangle$ is the dot product of two vectors of size k :

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle := \sum_{f=1}^k v_{i,f} \cdot v_{j,f} \quad (3)$$

A row $\mathbf{v}_i$ within $\mathbf{V}$ describes the i -th variable with k factors.

$k \in \mathbb{N}^+$ is a hyperparameter that defines the dimensionality of the factorization.

A 2-way FM (degree $d = 2$) captures all single and pairwise interactions between variables:

(1)의 두번째 항

(1)의 세번째 항

출처 : S. Rendle, IEEE-FM(2010)

1. Shallow와 Deep의 융합 - FM

즉, feature간의 관계와 중요성을 모델링합니다.

다만 계산 효율성을 고려해 2-way만 주로 사용합니다.

(이론상으로 d-way가 가능하지만, 계산 효율성이 떨어집니다.)

그래서 저차 상호작용(Low order interaction) 방법으로 많이 소개되거나

2차 상호작용(Second-order interaction)으로도 언급됩니다.

1. Shallow와 Deep의 융합

고차 상호작용(High order interaction)은 어떻게 고려할 수 있을까요?

즉! Shallow와 Deep을 결합해서 활용할 수는 없을까요?

다양한 상호작용을 고려할 수 있으면

좋은 효과를 낼 수 있지 않을까요?

1. Shallow와 Deep의 융합 - DeepFM

DeepFM: A Factorization-Machine based Neural Network for CTR Prediction

Huifeng Guo^{*1}, Ruiming Tang², Yunming Ye^{†1}, Zhenguo Li², Xiuqiang He²

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[출처 : H. Huo, IJCA DeepFM\(2017\)](#)

제목에서 볼 수 있듯

“뉴럴 네트워크 기반의 FM”

입니다.

1. Shallow와 Deep의 융합 - DeepFM

Abstract

Learning sophisticated feature interactions behind user behaviors is critical in maximizing CTR for recommender systems. Despite great progress, existing methods seem to have a strong bias towards low- or high-order interactions, or require expertise feature engineering. In this paper, we show that it is possible to derive an end-to-end learning model that emphasizes both low- and high-order feature interactions. The proposed model, DeepFM, combines the power of factorization machines for recommendation and deep learning for feature learning in a new neural network architecture. Compared to the latest Wide & Deep model from Google, DeepFM has a shared input to its “wide” and “deep” parts, with no need of feature engineering besides raw features. Comprehensive experiments are conducted to demonstrate the effectiveness and efficiency of DeepFM over the existing models for CTR prediction, on both benchmark data and commercial data.

출처 : H. Huo, IJCA_DeepFM(2017)

- 기존 연구들은 Low 또는 High 한쪽에 지나치게 편중되어 있음
- DeepFM은
 - Low와 High order feature interaction 고려
 - 두 개의 중요성을 결합하고 강조
- Factorization Machine(FM)과 뉴럴 네트워크를 결합
- Wide & Deep
 - Wide 파트를 FM으로 대체
 - 피처 엔지니어링이 필요 없음

1. Shallow와 Deep의 융합 - DeepFM

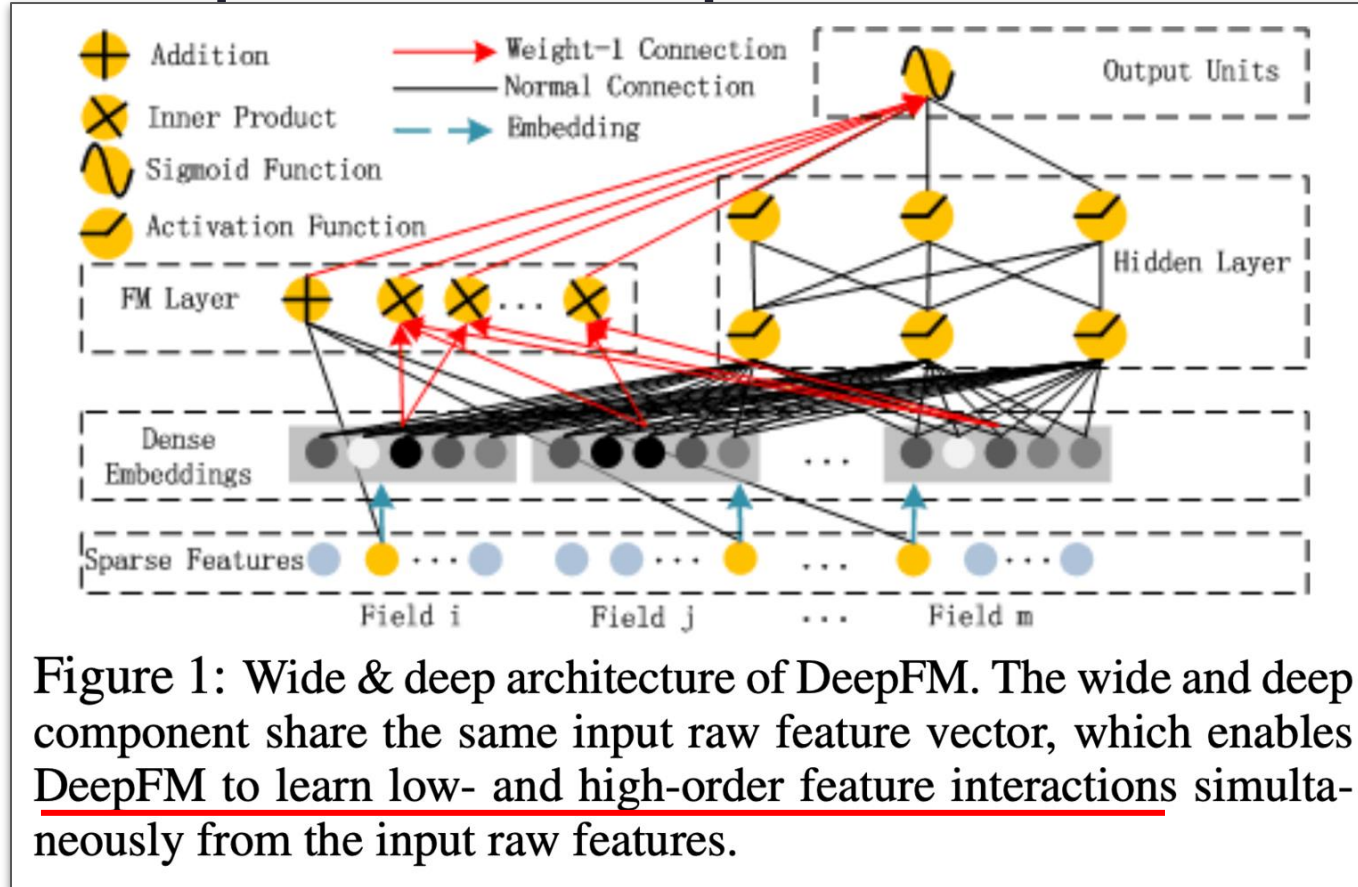


Figure 1: Wide & deep architecture of DeepFM. The wide and deep component share the same input raw feature vector, which enables DeepFM to learn low- and high-order feature interactions simultaneously from the input raw features.

출처 : H. Huo, IJCAI_DeepFM(2017)

1. Shallow와 Deep의 융합 - DeepFM

FM Component

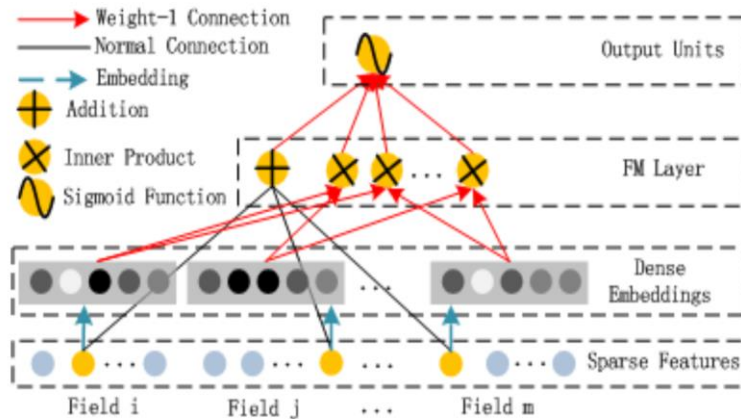


Figure 2: The architecture of FM.

The FM component is a factorization machine, which is proposed in [Rendle, 2010] to learn feature interactions for recommendation. Besides a linear (order-1) interactions among features, FM models pairwise (order-2) feature interactions as inner product of respective feature latent vectors.

출처 : H. Huo, IJCAI_DeepFM(2017)

Deep Component

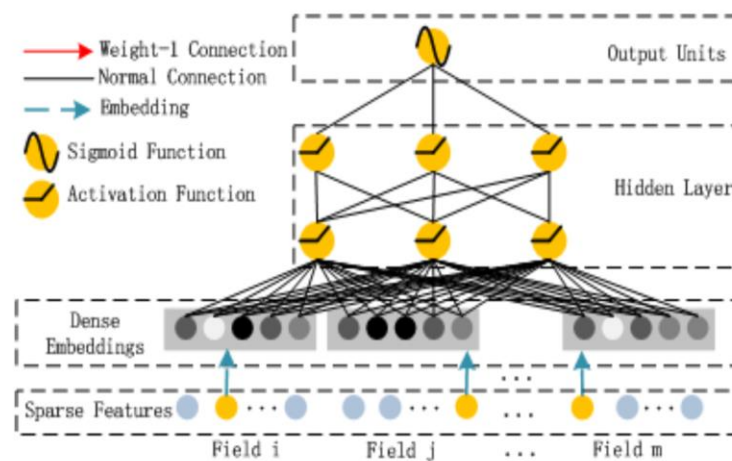


Figure 3: The architecture of DNN.

The deep component is a feed-forward neural network, which is used to learn high-order feature interactions. As shown in Figure 3, a data record (a vector) is fed into the neural network. Compared to neural networks with image [He

1. Shallow와 Deep의 융합 - DeepFM

FM Component

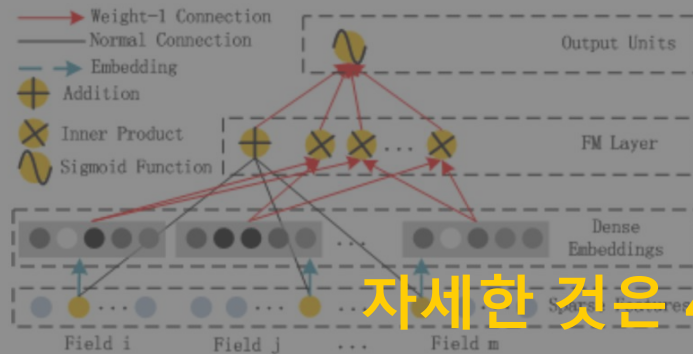


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Deep Component



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The deep component is a feed-forward neural network, which is used to learn high-order feature interactions. As shown in Figure 3, a data record (a vector) is fed into the neural network. Compared to neural networks with image [He

자세한 것은 4장에서 학습하겠습니다.

출처 : H. Huo, IJCAI_deepfm(2017)

1. Shallow와 Deep의 융합 - W&D

Wide & Deep Learning for Recommender Systems

Heng-Tze Cheng, Levent Koc, Jeremiah Harmsen, Tal Shaked, Tushar Chandra,
Hrishi Aradhye, Glen Anderson, Greg Corrado, Wei Chai, Mustafa Ispir, Rohan Anil,
Zakaria Haque, Lichan Hong, Vihan Jain, Xiaobing Liu, Hemal Shah
*
Google Inc.

[출처 : H. Cheng, acm_W&D\(2016\)](#)

구글에서 공개한 추천 시스템

“Wide part와 Deep part를 결합”

실제 google play에서 실험했던 논문입니다.

1. Shallow와 Deep의 융합 - W&D

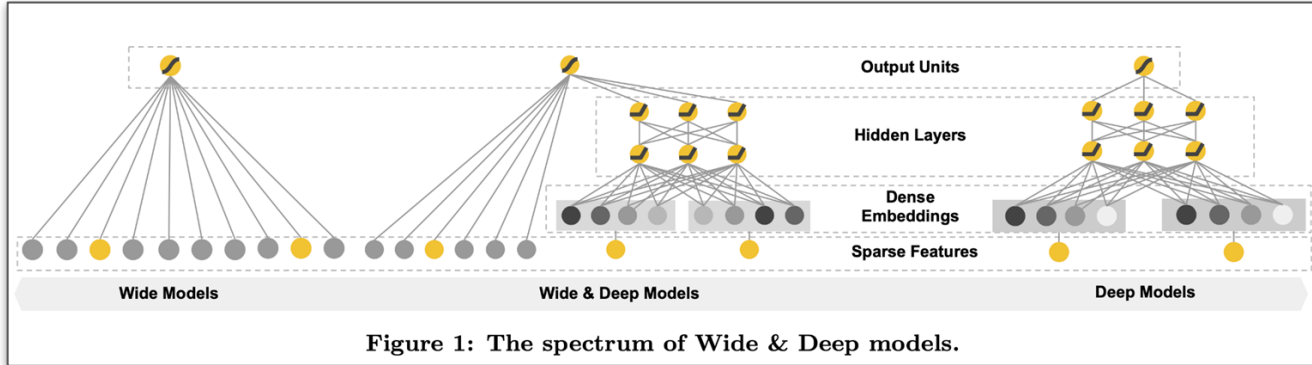
ABSTRACT

Generalized linear models with nonlinear feature transformations are widely used for large-scale regression and classification problems with sparse inputs. Memorization of feature interactions through a wide set of cross-product feature transformations are effective and interpretable, while generalization requires more feature engineering effort. With less feature engineering, deep neural networks can generalize better to unseen feature combinations through low-dimensional dense embeddings learned for the sparse features. However, deep neural networks with embeddings can over-generalize and recommend less relevant items when the user-item interactions are sparse and high-rank. In this paper, we present Wide & Deep learning—jointly trained wide linear models and deep neural networks—to combine the benefits of memorization and generalization for recommender systems. We productionized and evaluated the system on Google Play, a commercial mobile app store with over one billion active users and over one million apps. Online experiment results show that Wide & Deep significantly increased app acquisitions compared with wide-only and deep-only models. We have also open-sourced our implementation in TensorFlow.

출처 : [H. Cheng, acm_W&D\(2016\)](#)

- 기억 파트(Memorization)
 - Wide 파트를 담당
 - Feature간의 관계를 기억
 - 그러나 피처 엔지니어링 노력이 들어감
- 일반화 파트(Generalization)
 - Deep 뉴럴 네트워크를 활용
 - 피처 엔지니어링 노력이 적음
 - 그러나 과도하게 일반화될 수 있음
- Wide & Deep
 - 기억 파트와 일반화 파트 결합
 - 서로간의 장점을 결합

1. Shallow와 Deep의 융합 - W&D



3.3 Joint Training of Wide & Deep Model

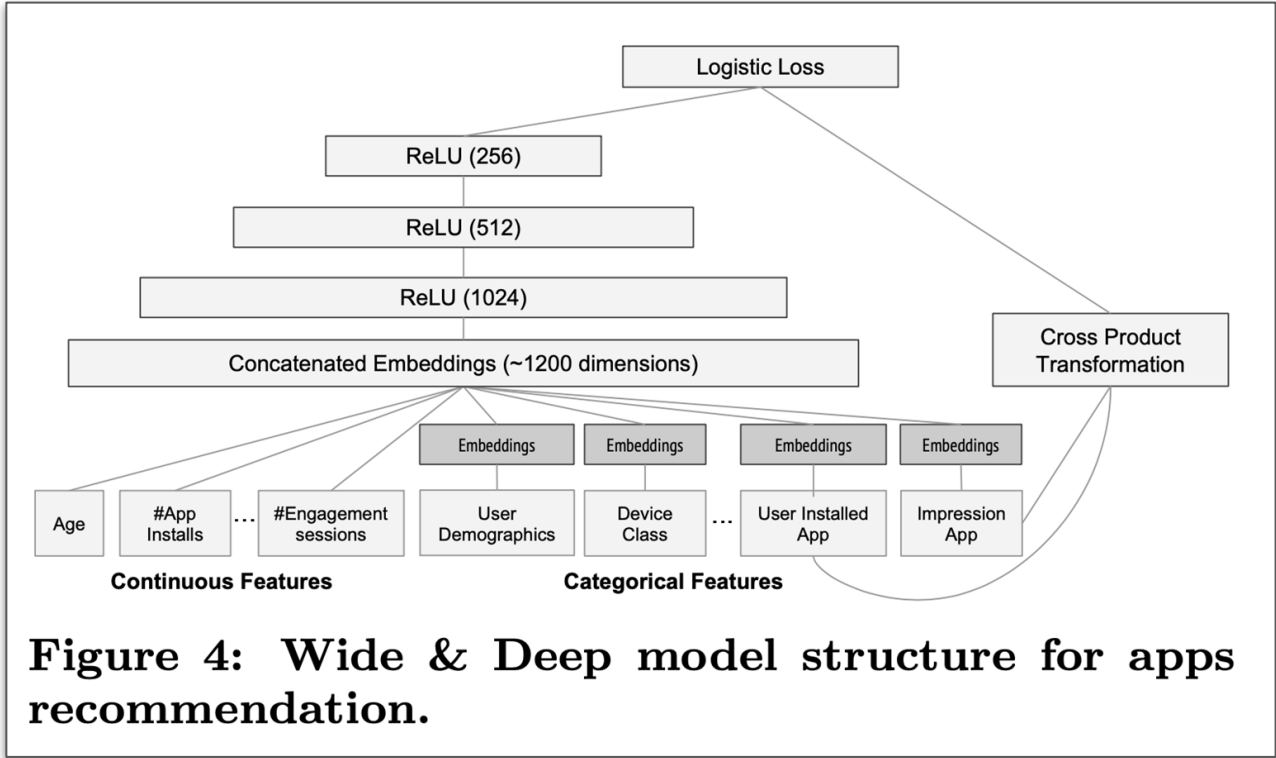
The wide component and deep component are combined using a weighted sum of their output log odds as the pre-

출처 : H. Cheng, acm_W&D(2016)

Joint training과 Ensemble의 차이

- Ensemble: 개별 모델들이 독립적으로 훈련하며 Inference 때에만 결합
- **Joint training**: 공동으로 모델을 훈련하면서 가중치들을 최적화

1. Shallow와 Deep의 융합 - W&D



출처 : [H. Cheng, acm_W&D\(2016\)](#)

Table 1: Offline & online metrics of different models. Online Acquisition Gain is relative to the control.		
Model	Offline AUC	Online Acquisition Gain
Wide (control)	0.726	0%
Deep	0.722	+2.9%
Wide & Deep	0.728	+3.9%

1. Shallow와 Deep의 융합

Shallow와 Deep을 융합하는 방법은
지금도 활발하게 연구가 되고 있습니다.

서로 장점을 보완해주는 역할로 작용하며
좋은 성능과 연구 성과를 보여주고 있습니다.

03

어텐션을 활용한 추천 시스템

1. 어텐션

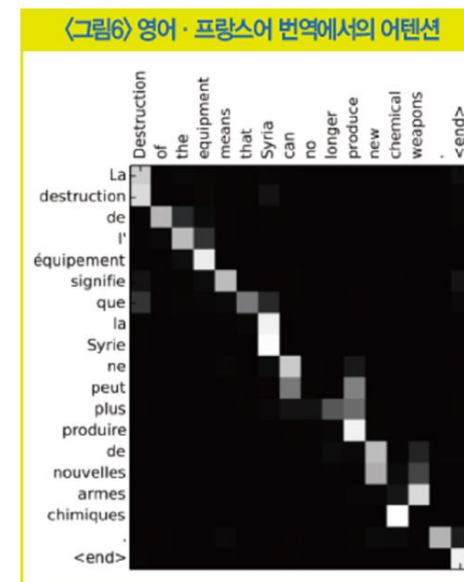
2. 어텐션과 추천 시스템

1. 어텐션

어텐션 메커니즘(Attention Mechanism)
들어보셨나요?

1. 어텐션

- 집중!(Attention)
 - 뉴럴 네트워크가 입력 데이터의 특정 부분에 집중할 수 있는 방법
 - 마치 사람이 사물을 볼 때 특정 그림, 키워드 등에 집중하는 것과 유사
 - 가장 관련있는 정보에 집중하여 정보를 효율적으로 처리 및 전달
 - 번역할 때 가장 도움이 되는, 연관성 있는 단어를 활용해 번역 수행
- 다양한 분야에서 사용됩니다.
 - Self-attention, Transformer 계열로 인해 NLP에서 특히 유명하죠!
 - 이미지쪽에서도 굉장히 잘 사용됩니다.
 - 추천 시스템에서도 활용될 수 있습니다



출처 : 한경매거진

1. 어텐션

**어텐션 메커니즘(Attention Mechanism)에는
다양한 어텐션 방법들이 있습니다.**

1. 어텐션 - Bahdanau

Published as a conference paper at ICLR 2015

NEURAL MACHINE TRANSLATION BY JOINTLY LEARNING TO ALIGN AND TRANSLATE

Dzmitry Bahdanau
Jacobs University Bremen, Germany

KyungHyun Cho **Yoshua Bengio***
Université de Montréal

[출처 : Bahdanau, ICRL-Bahdanau_att\(2015\)](#)

Bahdanau 어텐션으로 유명한 논문

- 기본적인 Attention으로 많이 언급되고 있음
- Encoder - Decoder 구조에서 활용

1. 어텐션 - Bahdanau

3.1 DECODER: GENERAL DESCRIPTION

In a new model architecture, we define each conditional probability in Eq. (2) as:

$$p(y_i|y_1, \dots, y_{i-1}, \mathbf{x}) = g(y_{i-1}, s_i, c_i), \quad (4)$$

where s_i is an RNN hidden state for time i , computed by

$$s_i = f(s_{i-1}, y_{i-1}, c_i).$$

It should be noted that unlike the existing encoder-decoder approach (see Eq. (2)), here the probability is conditioned on a distinct context vector c_i for each target word y_i .

The context vector c_i depends on a sequence of *annotations* ($h_1, \dots, h_{T_x}$) to which an encoder maps the input sentence. Each annotation h_i contains information about the whole input sequence with a strong focus on the parts surrounding the i -th word of the input sequence. We explain in detail how the annotations are computed in the next section.

The context vector c_i is, then, computed as a weighted sum of these annotations h_i :

$$c_i = \sum_{j=1}^{T_x} \alpha_{ij} h_j. \quad (5)$$

The weight α_{ij} of each annotation h_j is computed by

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})}, \quad (6)$$

where

$$e_{ij} = a(s_{i-1}, h_j)$$

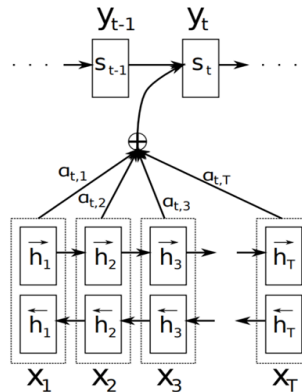


Figure 1: The graphical illustration of the proposed model trying to generate the t -th target word y_t given a source sentence $(x_1, x_2, \dots, x_T)$.

Decoder에 Attention을 적용

- Encoder의 정보를 활용함
- Softmax 계산으로 Attention Score 추출
- 추출된 Score를 Decoder에서 사용
- 가중합으로 정보 압축

1. 어텐션 - Global & Local

Effective Approaches to Attention-based Neural Machine Translation

Minh-Thang Luong Hieu Pham Christopher D. Manning
Computer Science Department, Stanford University, Stanford, CA 94305
{lmthang, hyhieu, manning}@stanford.edu

출처 : [M. Luong, EMNLP att\(2015\)](#)

- Bahdanau 어텐션을 좀 더 보강한 후속 논문
- **Dot product attention**으로도 많이 언급됨
- **Global attention** 및 **Local attention**으로 유명한 논문

1. 어텐션 - Global & Local

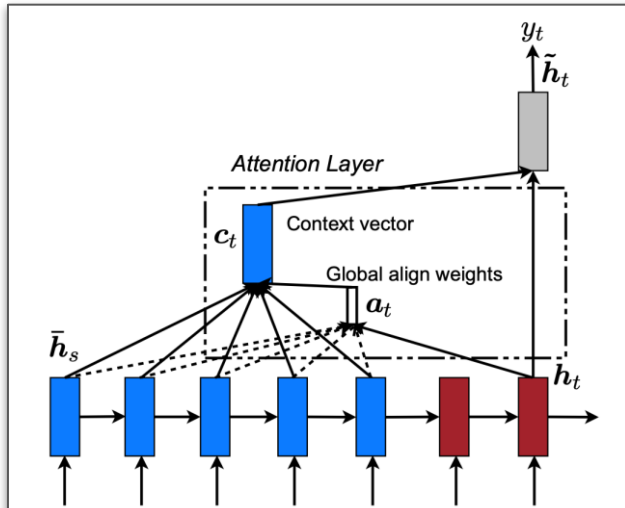


Figure 2: **Global attentional model** – at each time step t , the model infers a *variable-length* alignment weight vector a_t based on the current target state h_t and all source states $\bar{h}_s$. A global context vector c_t is then computed as the weighted average, according to a_t , over all the source states.

출처 : M. Luong, EMNLP_att(2015)

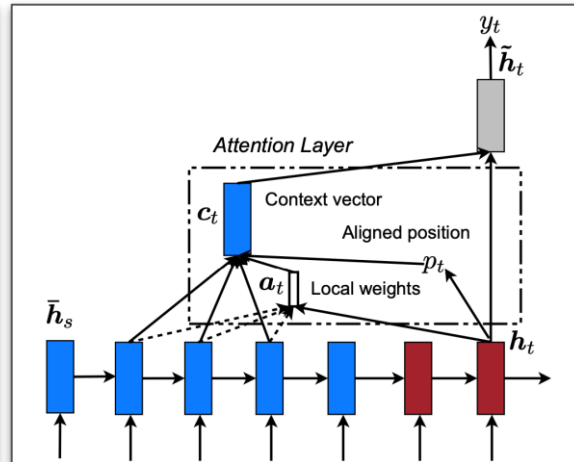


Figure 3: **Local attention model** – the model first predicts a single aligned position p_t for the current target word. A window centered around the source position p_t is then used to compute a context vector c_t , a weighted average of the source hidden states in the window. The weights a_t are inferred from the current target state h_t and those source states $\bar{h}_s$ in the window.

Global attention 및 Local attention

- Global
 - Input sequence 전체에 대해 정보 고려
 - 연산량이 많아질 수 있음
- Local
 - 특정 Subset 기준으로 Attention 고려
 - 정보가 한정적일 수 있음

1. 어텐션 - Transformer

Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com	Noam Shazeer* Google Brain noam@google.com	Niki Parmar* Google Research nikip@google.com	Jakob Uszkoreit* Google Research usz@google.com
Llion Jones* Google Research llion@google.com	Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu	Łukasz Kaiser* Google Brain lukaszkaizer@google.com	
Illia Polosukhin* ‡ illia.polosukhin@gmail.com			

출처 : Vaswani, NIPS_Transformer(2017)

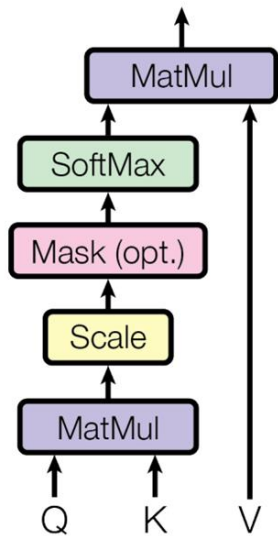
너무너무너무너무너무 유명한 논문

- Transformer라고 언급되는 논문
- BERT, GPT 계열 모델의 전신이 되는 논문

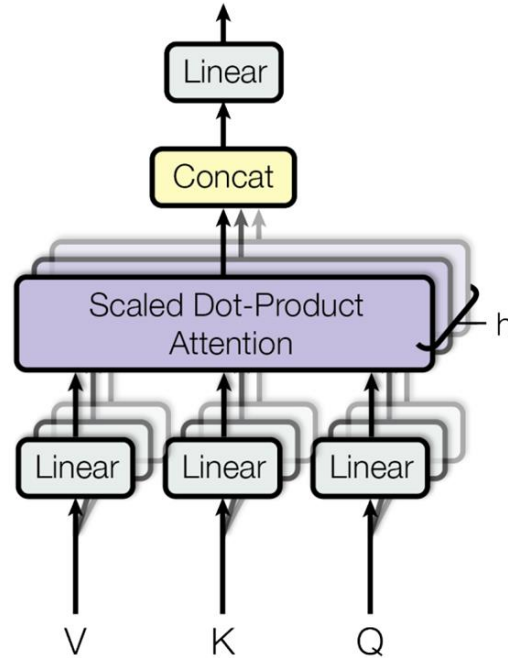
Self-Attention 방법으로 유명

1. 어텐션 - Transformer

Scaled Dot-Product Attention



Multi-Head Attention

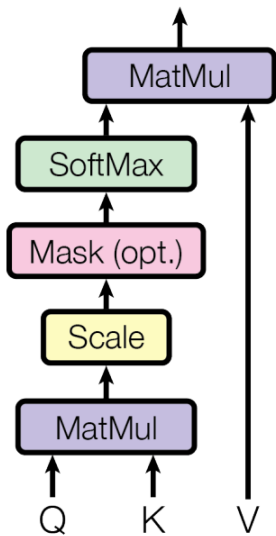


- Self-Attention으로 유명
- Multi-Head Self-Attention 구조 도입

출처 : [Vaswani, NIPS_Transformer\(2017\)](#)

1. 어텐션 - Transformer

Scaled Dot-Product Attention



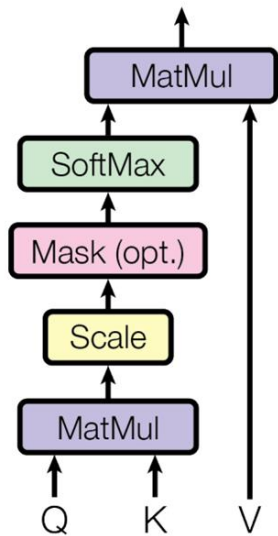
[0.0012, 0.0034, 0.0078]

[0.1, 0.7, 0.2]

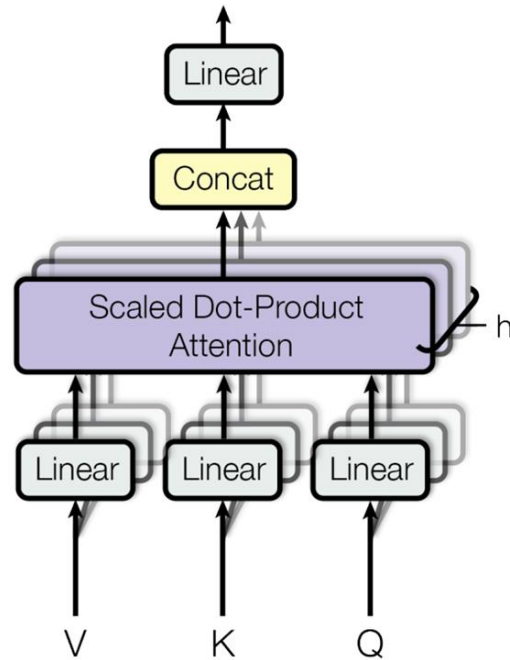
출처 : [Vaswani, NIPS_Transformer\(2017\)](#)

1. 어텐션 - Transformer

Scaled Dot-Product Attention



Multi-Head Attention



- Self-Attention으로 유명
- Multi-Head Self-Attention 구조 도입

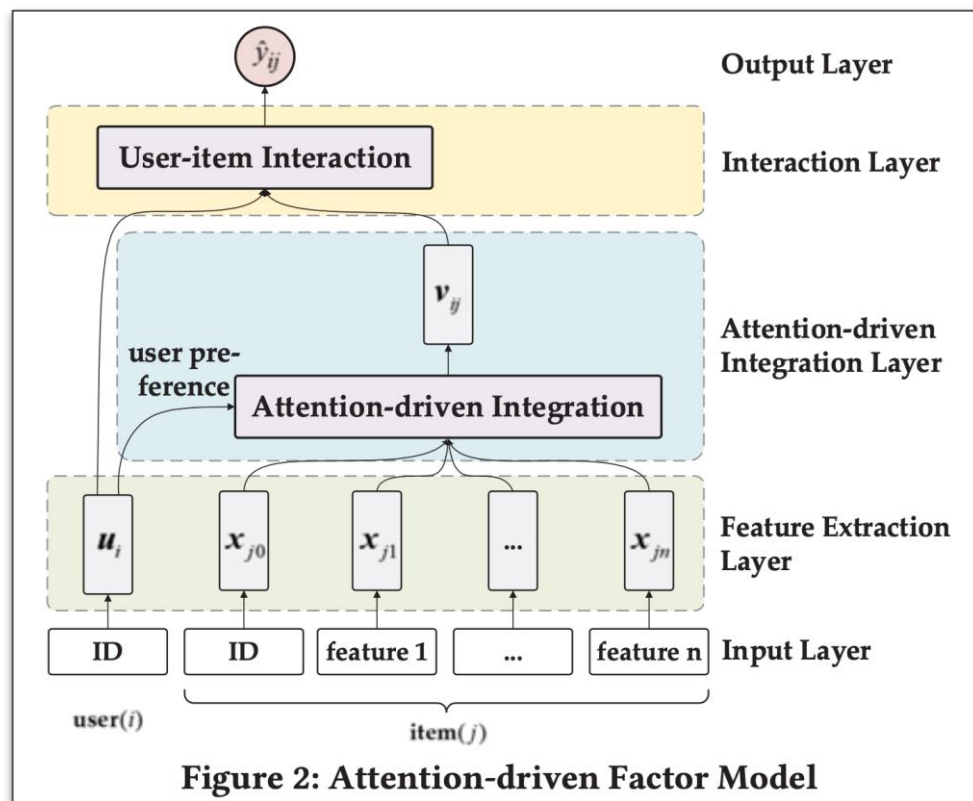
출처 : [Vaswani, NIPS_Transformer\(2017\)](#)

2. 어텐션과 추천 시스템

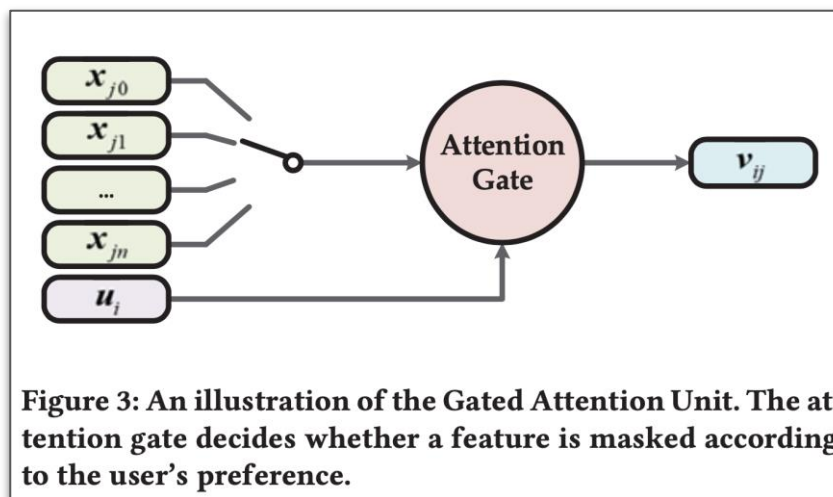
어텐션 메커니즘(Attention Mechanism)을 쓰는 이유는
“중요한 것”에 집중하기 위함입니다.

추천 시스템에서는 어떻게 활용할 수 있을까요?

2. 어텐션과 추천 시스템



출처 : J. Chen, acm_afm(2018)



- 사용자의 “why(왜 상호작용을 했는가)”에 대한 답을 도울 수 있음
- 다양한 아이템 피처에 대한 사용자의 어텐션 분포를 추정
- 아이템 피처에 대해서 일종의 가중치를 다르게 부여할 수 있음

2. 어텐션과 추천 시스템

Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI-19)

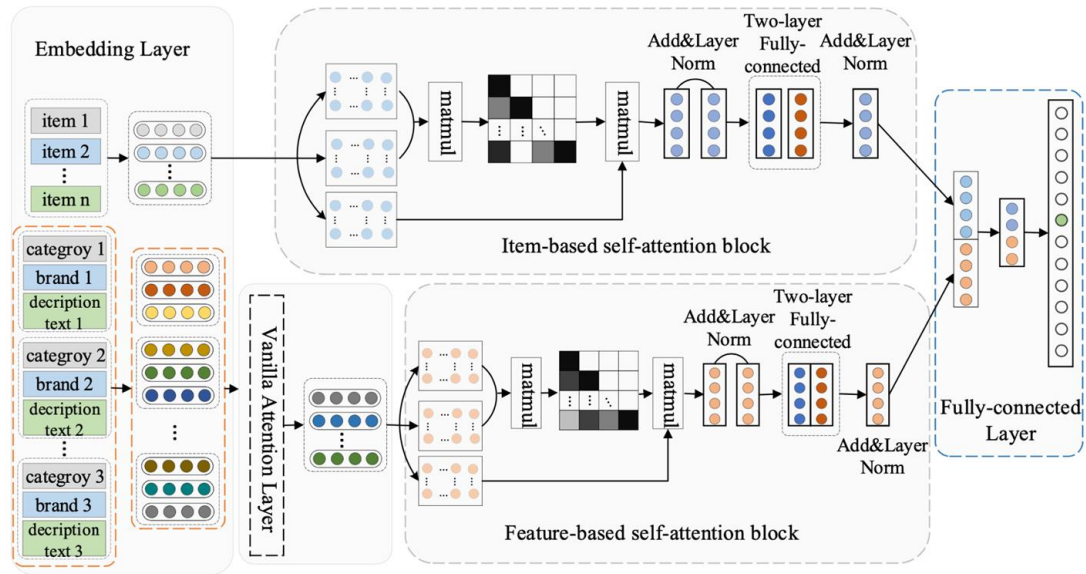


Figure 1: The Network Architecture of FDSA.

출처 : [Zhang, ijcai_FDSA\(2019\)](#)

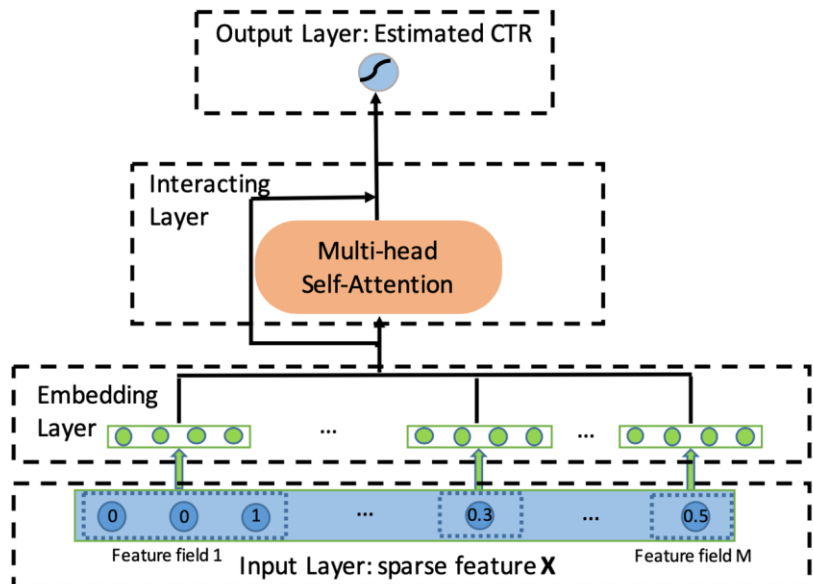


Figure 1: Overview of our proposed model AutoInt. The details of embedding layer and interacting layer are illustrated in Figure 2 and Figure 3 respectively.

출처 : [Song, cikm_AutoInt\(2019\)](#)

2. 어텐션과 추천 시스템

AutoInt: Automatic Feature Interaction Learning via Self-Attentive Neural Networks

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출처 : [Song, cikm_AutoInt\(2019\)](#)

추후 프로젝트에서 사용할 연구입니
다.

(4장에서 자세히 배워볼 겁니다.)

제목에서 보이듯

“피쳐 상호작용을 셀프 어텐션으로 학습”

합니다.

2. 어텐션과 추천 시스템

추천 시스템에서 활용하는 어텐션 방법은
추천 결과(CTR 등)에 영향을 주는 Feature의 중요성을
학습하는 등 다양하게 활용되고 있습니다.

04

다양한 딥러닝 기반 추천 시스템

1. Sequential

2. Graph

3. Image

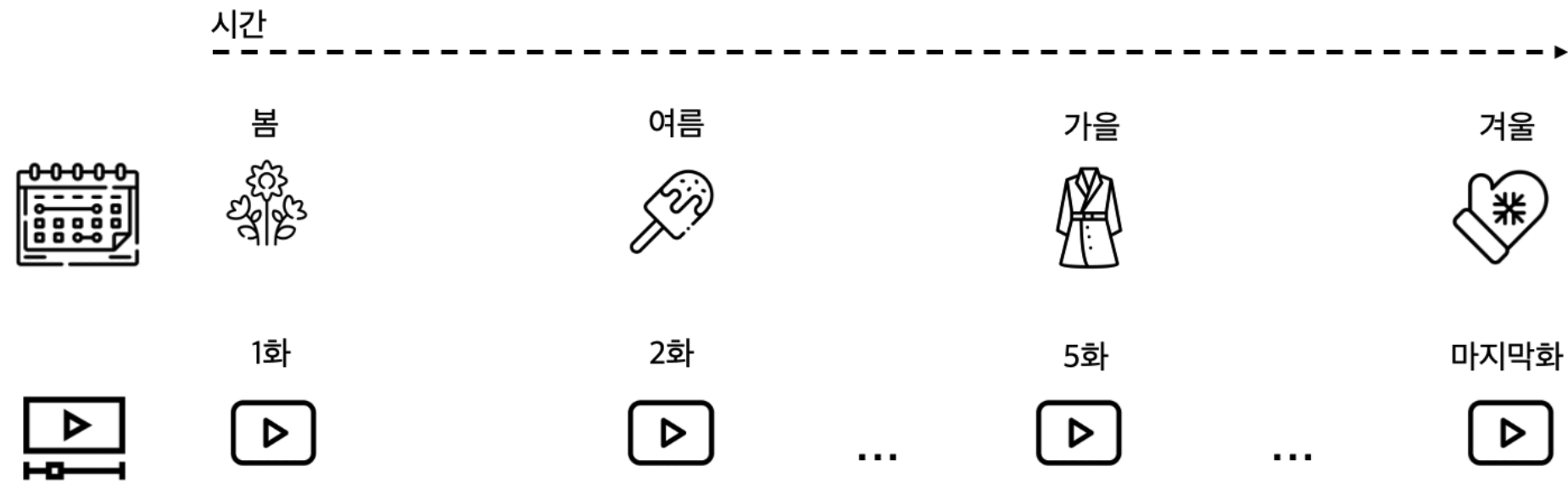
4. LLM

1. Sequential

살다보면 ‘**주기**’별로 어떤 행동이
일어날(발생할) 때가 있습니다.

아니면, 어떤 ‘**순서**’대로 해야할 것이 있죠.

1. Sequential



1. Sequential

‘하루’ 기준으로 아침, 점심, 저녁

‘주’ 단위로 월화수목금토일

‘월’ 단위로 ...

음악을 들을 때

쇼핑할 때

강의를 들을 때

수많은 ‘Sequential’ 패턴이 존재합니다.

1. Sequential - GRU4Rec

SESSION-BASED RECOMMENDATIONS WITH RECURRENT NEURAL NETWORKS

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ABSTRACT

We apply recurrent neural networks (RNN) on a new domain, namely recommender systems. Real-life recommender systems often face the problem of having to base recommendations only on short session-based data (e.g. a small sportware website) instead of long user histories (as in the case of Netflix). In this situation the frequently praised matrix factorization approaches are not accurate. This problem is usually overcome in practice by resorting to item-to-item recommendations, i.e. recommending similar items. We argue that by modeling the whole session, more accurate recommendations can be provided. We therefore propose an RNN-based approach for session-based recommendations. Our approach also considers practical aspects of the task and introduces several modifications to classic RNNs such as a ranking loss function that make it more viable for this specific problem. Experimental results on two data-sets show marked improvements over widely used approaches.

출처 : [Hidasi, arxiv GRU4Rec\(2016\)](#)

- RNN을 추천 시스템에 적용한 유명한 논문
- 기존에는 짧은 히스토리에 집중해 모델링 방법을 연구함
 - 본 논문은 RNN을 활용해 좀 더 긴 History에 적용함
- 순서를 모델링해야하는 상황에서 Matrix Factorization 같은 방법은 적절하지 않음
- 본 논문은 RNN-based(GRU) 접근 방법으로 세션을 모델링

1. Sequential - SASRec

Self-Attentive Sequential Recommendation

Wang-Cheng Kang, Julian McAuley
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Abstract—Sequential dynamics are a key feature of many modern recommender systems, which seek to capture the ‘context’ of users’ activities on the basis of actions they have performed recently. To capture such patterns, two approaches have proliferated: Markov Chains (MCs) and Recurrent Neural Networks (RNNs). Markov Chains assume that a user’s next action can be predicted on the basis of just their last (or last few) actions, while RNNs in principle allow for longer-term semantics to be uncovered. Generally speaking, MC-based methods perform best in extremely sparse datasets, where model parsimony is critical, while RNNs perform better in denser datasets where higher model complexity is affordable. The goal of our work is to balance these two goals, by proposing a self-attention based sequential model (SASRec) that allows us to capture long-term semantics (like an RNN), but, using an attention mechanism, makes its predictions based on relatively few actions (like an MC). At each time step, SASRec seeks to identify which items are ‘relevant’ from a user’s action history, and use them to predict the next item. Extensive empirical studies show that our method outperforms various state-of-the-art sequential models (including MC/CNN/RNN-based approaches) on both sparse and dense datasets. Moreover, the model is an order of magnitude more efficient than comparable CNN/RNN-based models. Visualizations on attention weights also show how our model adaptively

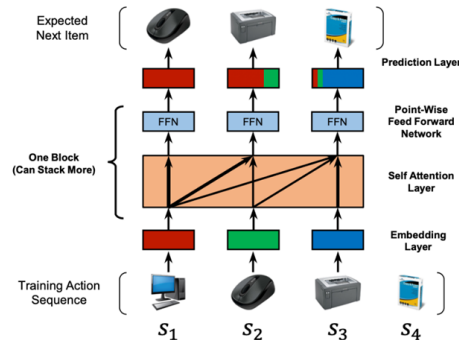


Figure 1: A simplified diagram showing the training process of SASRec. At each time step, the model considers all previous items, and uses attention to ‘focus on’ items relevant to the next action.

- 셀프 어텐션 기반의 Sequential 모델 구조를 활용
- RNN처럼 장기적 의미를 파악할 수 있고, 어텐션을 활용해 상대적으로 ‘적은’ 사용자 히스토리 기반으로 예측을 수행할 수 있는 SASRec 제안
- 마르코프 기반 모델과 RNN 기반 모델보다 훨씬 효율적이고 성능도 뛰어남

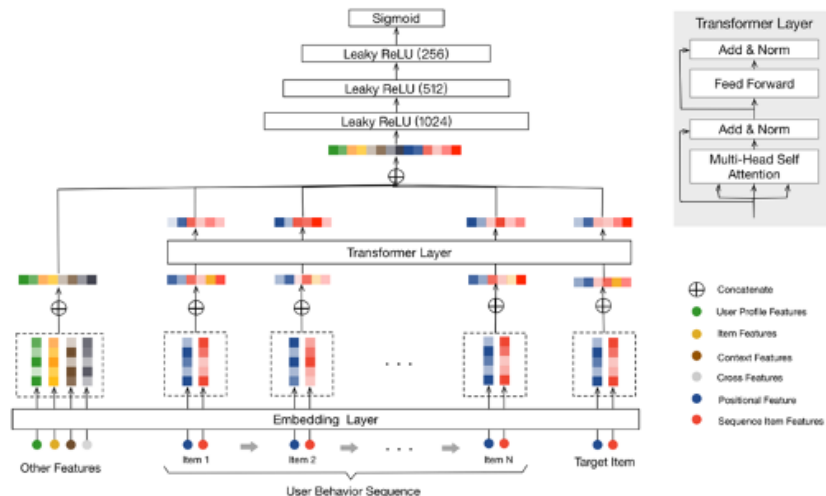
출처 : Kang, arxiv_SASRec(2018)

1. Sequential - BST

Behavior Sequence Transformer for E-commerce Recommendation in Alibaba

Qiwei Chen, Huan Zhao*
Wei Li, Pipei Huang, Wenwu Ou
Alibaba Search&Recommendation Group
Beijing&Hangzhou, China

{chenqiwei.cqw,chuanfei.zh,rob.lw,pipei.hpp,santong.oww}@alibaba-inc.com



출처 : [Chen, acm BST\(2019\)](#)

- 알리바바에서 제안한 논문
- Transformer 구조 기반의 Sequence base 모델
- 사용자의 순차적 신호를 캡처하여 행동 시퀀스를 모델링
 - 사용자1 : $A \rightarrow B \rightarrow C \rightarrow F$
 - 사용자2 : $B \rightarrow C \rightarrow F$

1. Sequential - BERT4Rec

BERT4Rec: Sequential Recommendation with Bidirectional Encoder Representations from Transformer

Fei Sun, Jun Liu, Jian Wu, Changhua Pei, Xiao Lin, Wenwu Ou, and Peng Jiang
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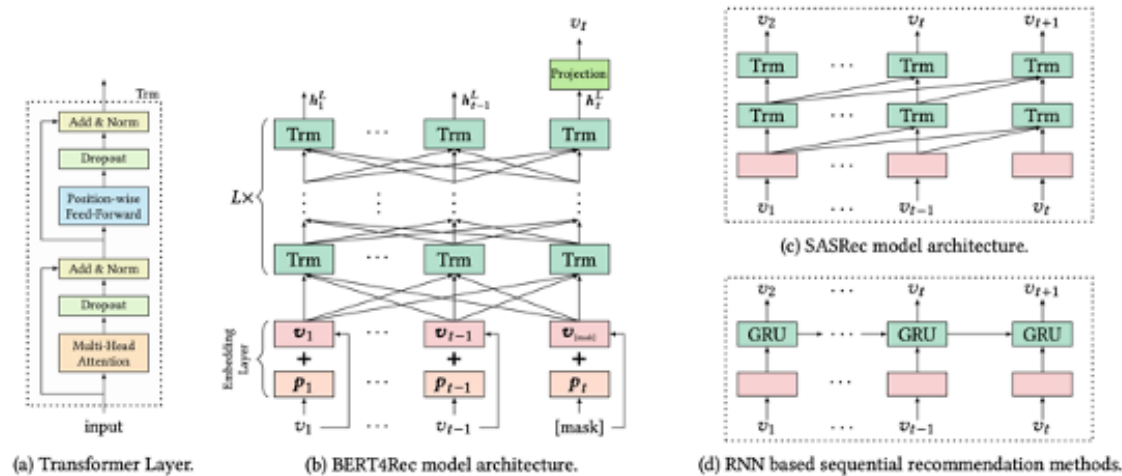


Figure 1: Differences in sequential recommendation model architectures. BERT4Rec learns a bidirectional model via Cloze task, while SASRec and RNN based methods are all left-to-right unidirectional model which predict next item sequentially.

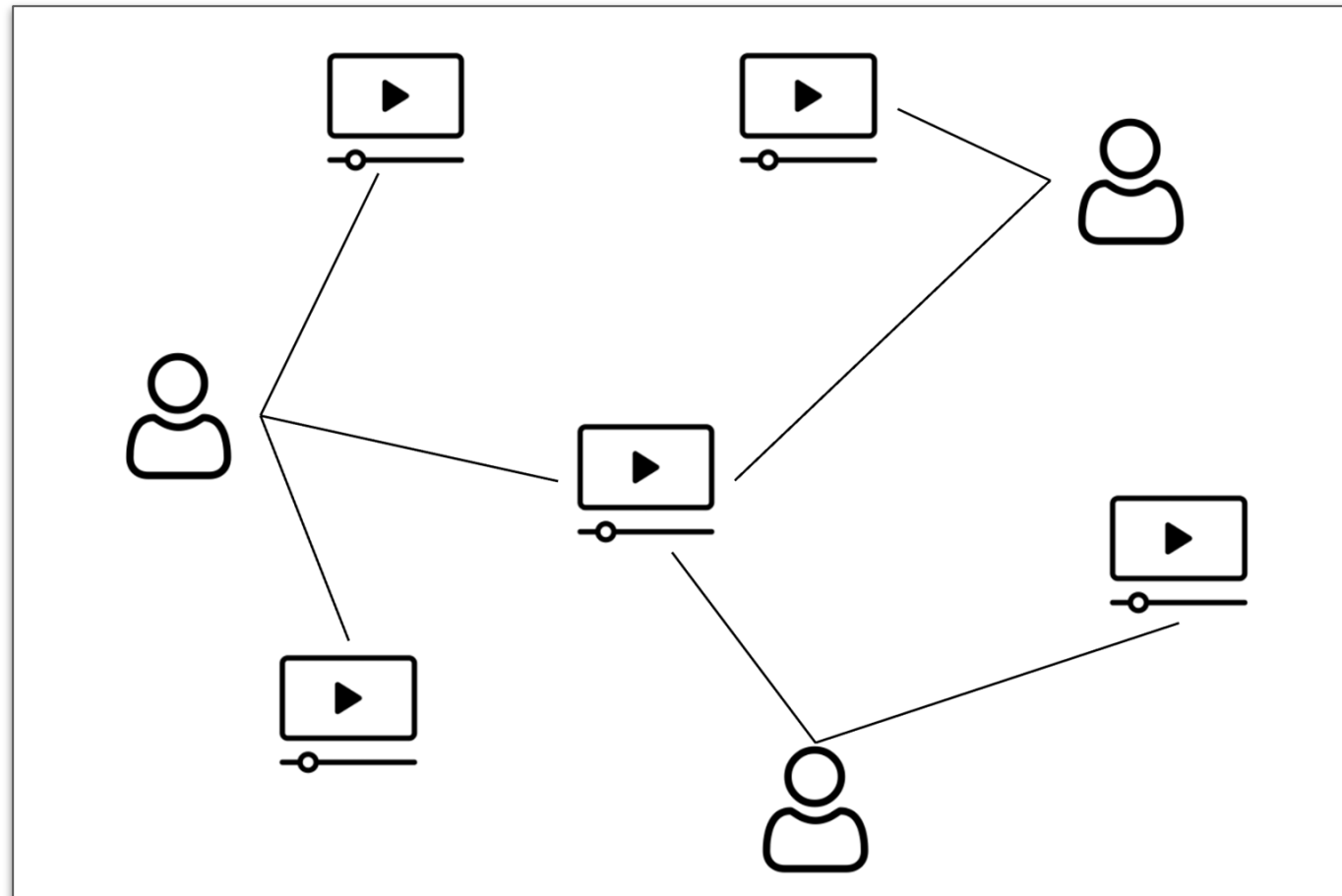
출처 : Fei Sun, arxiv_BERT4Rec(2019)

- 유명한 BERT 모델 구조를 추천 시스템에 적용한 연구
- 이전의 방법들은 사용자의 히스토리를 인코딩하기 위해 왼쪽에서 오른쪽으로 이어지는 순서 행동을 인코딩하였고
이때 시퀀셜 구조를 사용
 - 좋은 성능을 보여주었으나 한계가 존재
 - 단방향 아키텍처는 사용자 시퀀스의 잠재적 표현을 제한함
 - 항상 실용적이지 않은 정렬된 시퀀스를 가정
- BERT4Rec은 양방향(Bidirectional) 셀프 어텐션 구조를 활용
- 랜덤 마스크 아이템을 예측하도록 설계

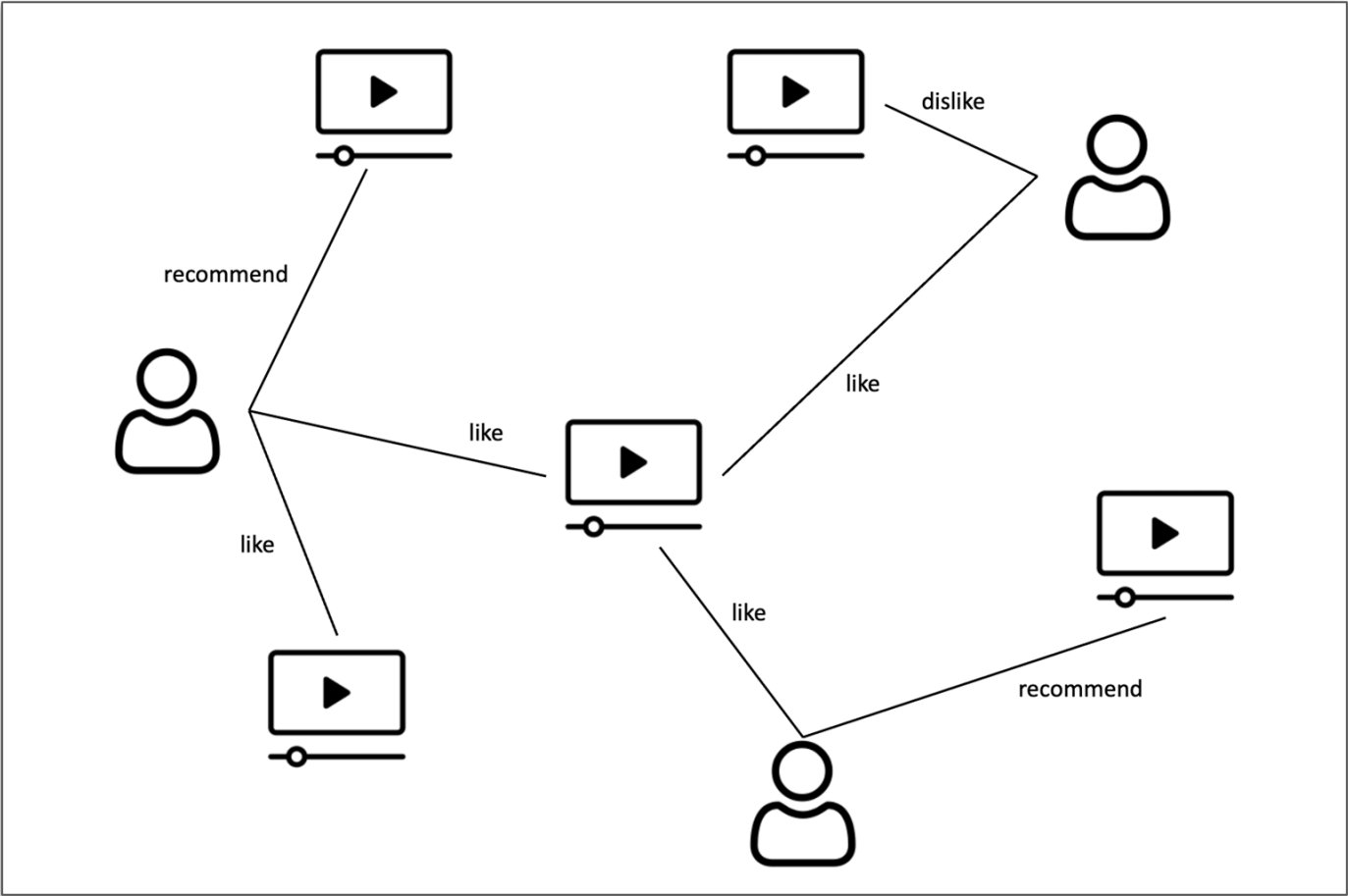
2. Graph

그래프란 무엇일까요?
“노드”와 “엣지(연결된 선)”으로
구성된 형태입니다.

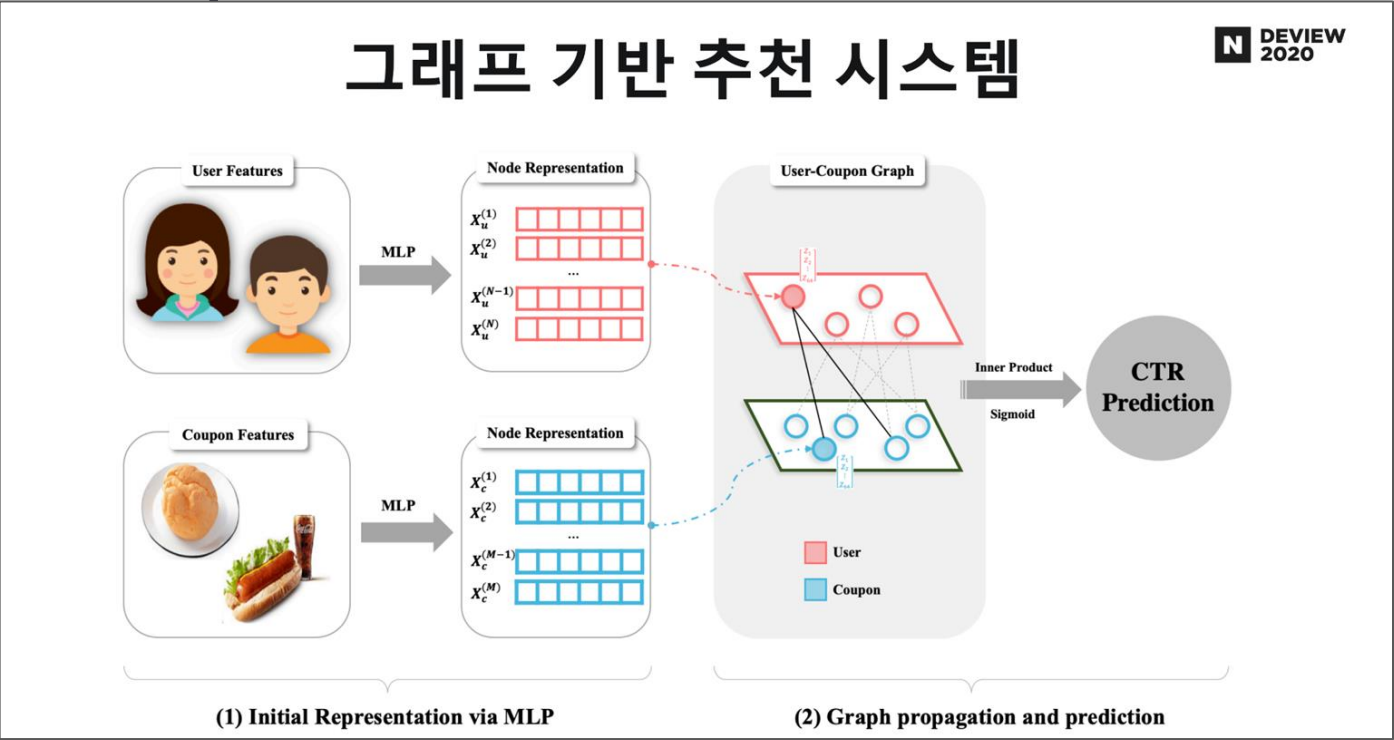
2. Graph



2. Graph



2. Graph



출처: <https://url.kr/zsu3mx>

DEVIEW 추천시스템3.0 자료에
서도
그래프 추천이 있었습니다.

2. Graph - NGCF

Neural Graph Collaborative Filtering*

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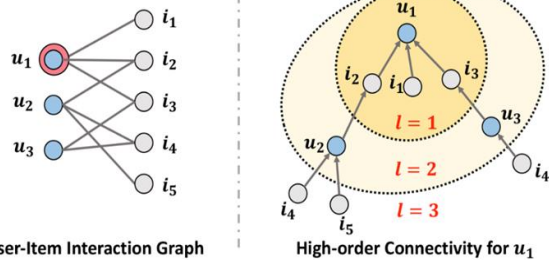


Figure 1: An illustration of the user-item interaction graph and the high-order connectivity. The node u_1 is the target user to provide recommendations for.

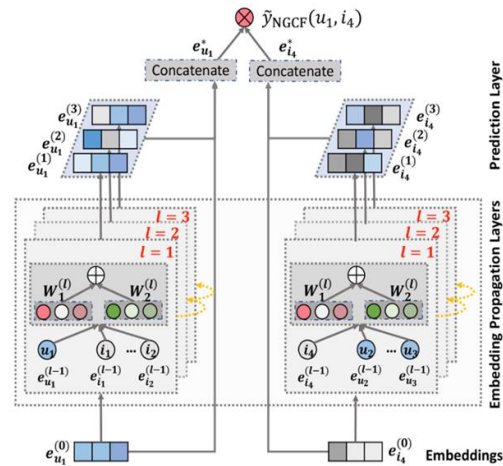


Figure 2: An illustration of NGCF model architecture (the arrowed lines present the flow of information). The representations of user u_1 (left) and item i_4 (right) are refined with multiple embedding propagation layers, whose outputs are concatenated to make the final prediction.

- CF에서 NCF로 넘어온 연구 방법을 그래프에 확장시킨 연구
- 기존 임베딩 방법들은 사용자-아이템 상호 작용을 효과적으로 인코딩하지 못하였음
 - 협업 필터링 효과를 캡처하기에 충분하지 않을 수 있음
- 사용자-아이템 상호작용을 효과적으로 인코딩할 수 있는 방법 제안
 - 이분 그래프(Bipartite graph) 구조
 - 사용자-아이템 그래프 구조
 - Neural Graph Collaborative Filtering(NGCF)

출처 : Wang, arxiv_NGCF(2019)

2. Graph - LightGCN

LightGCN: Simplifying and Powering Graph Convolution Network for Recommendation

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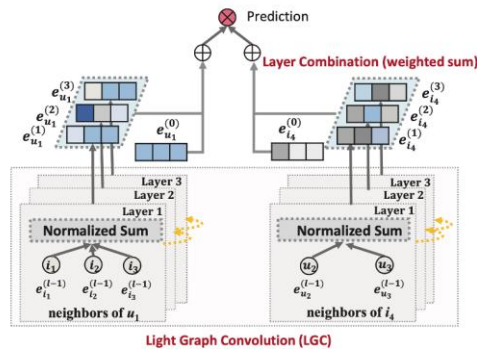


Figure 2: An illustration of LightGCN model architecture. In LGC, only the normalized sum of neighbor embeddings is performed towards next layer; other operations like self-connection, feature transformation, and nonlinear activation are all removed, which largely simplifies GCNs. In Layer Combination, we sum over the embeddings at each layer to obtain the final representations.

- Graph Convolution Network(GCN) 기반 방법을 추천 시스템에 맞게 조정하는 분석 과정이 부족하였음
- GCN에서 사용하는 Feature transformation과 비선형 활성화(Nonlinear activation) 함수가 협업 필터링 성능에 거의 기여하지 못함을 실험적으로 발견
- 본 논문에서는 GCN의 디자인을 단순화하여 추천 시스템에 적절하도록 변환
- NGCF보다 상당한 개선을 보임
- 불필요한 연산을 없애 가벼우면서 효과적인 추천 구조 제안

출처 : He, acm_LightGCN(2020)

3. Image

이미지를 이용한 추천 시스템 방법도 있습니다.

사용자가 선호하는 이미지 유형을 분석하거나,
비슷한 이미지 특징을 모아서 클러스터링하거나,
RNN과 같이 결합하는 등의
다양한 방법을 적용하고 있습니다.

3. Image

Image-based Recommendations on Styles and Substitutes

Julian McAuley^{*1}, Christopher Targett^{†2}, Qinfeng ('Javen') Shi^{‡2}, and Anton van den Hengel^{§2,3}

¹Department of Computer Science, UC San Diego

²School of Computer Science, University of Adelaide

³Australian Centre for Robot Vision

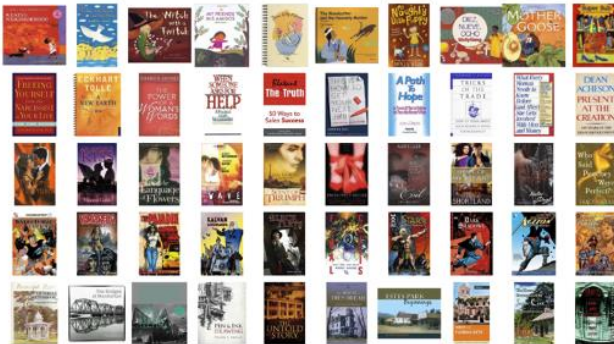


Figure 5: Examples of K-means clusters in style space (Books 'also viewed' and 'also bought' data). Although 'styles' for categories like books are not so readily interpretable as they are for clothes, visual features are nevertheless able to uncover meaningful distinctions between different product categories, e.g. the first four rows above appear to be children's books, self-help books, romance novels, and graphic novels.

- 사람은 어떤 사물을 볼 때 '시각적' 요소를 보게 되어 있음
- 어떤 물건들은 대체되거나 또는 다른 물건과 상호 보완적인 물건일 수 있음
 - 대체 : 청바지 두 벌
 - 상호보완 : 셔츠 한 장과 청바지 한 벌
- 사람과 시각적 관계 사이의 개념을 모델링

출처 : [McAuley, acm_img_rec1\(2015\)](#)

3. Image

Image-based Recommendations on Styles and Substitutes

Julian McAuley*¹, Christopher Targett^{1,2}, Qinfeng (‘Javen’) Shi^{1,2}, and Anton van den Hengel^{§2,3}

¹Department of Computer Science, UC San Diego
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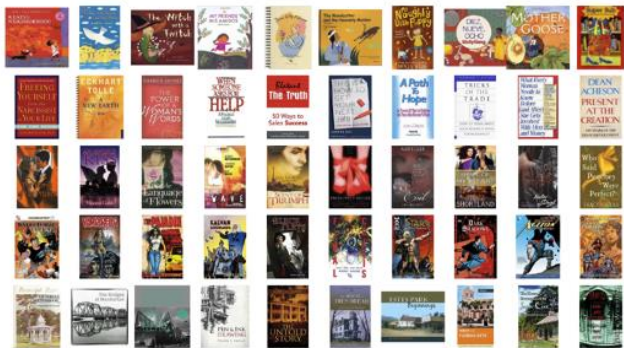


Figure 5: Examples of K-means clusters in style space (Books ‘also viewed’ and ‘also bought’ data). Although ‘styles’ for categories like books are not so readily interpretable as they are for clothes, visual features are nevertheless able to uncover meaningful distinctions between different product categories, e.g. the first four rows above appear to be children’s books, self-help books, romance novels, and graphic novels.



Figure 8: Outfits generated by our algorithm (Women’s outfits at left; Men’s outfits at right). The first column shows a ‘query’ item that is randomly selected from the product catalogue. The right three columns match the query item with a top, pants, shoes, and an accessory, (minus whichever category contains the query item).



Figure 7: A 2-dimensional embedding of a small sample of Boys clothing images (‘also viewed’ data).

출처 : McAuley, acm_img_rec1(2015)

3. Image - VECF

Personalized Fashion Recommendation with Visual Explanations based on Multimodal Attention Network

Towards Visually Explainable Recommendation

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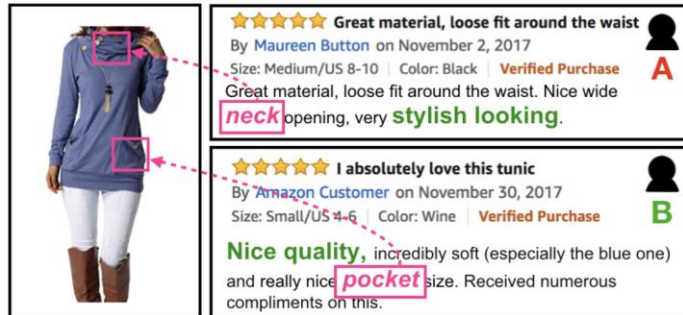
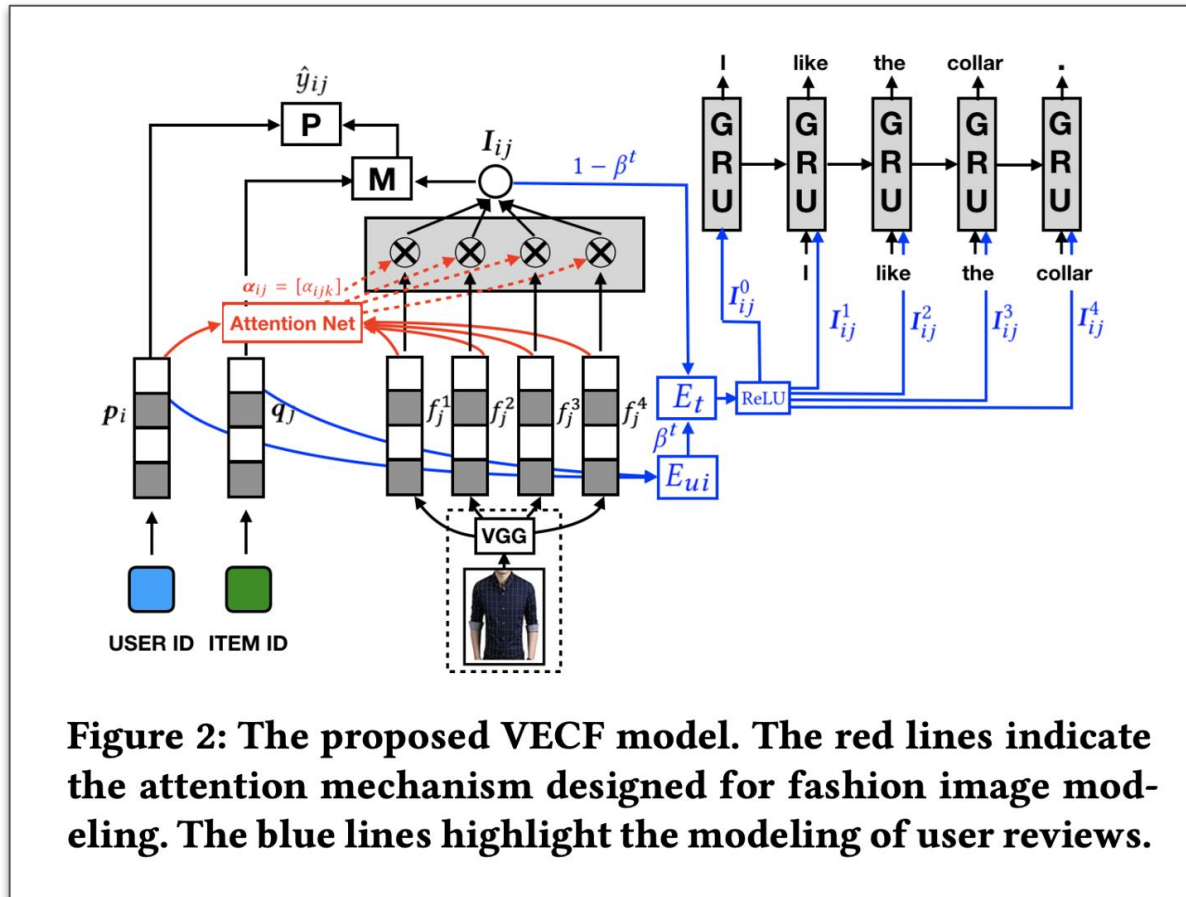


Figure 1: Example of users with different fine-grained visual preferences. User reviews may have partial correspondences to the fashion image. The pink italic and green bold fonts indicate the review information that can and cannot be aligned with some specific image regions.

출처 : Chen, acm_VECF(2019)

- 이미지 영역의 특징과 사용자 리뷰 정보를 기반으로 새로운 추천 아키텍처 제안
- 패션 이미지의 경우 모든 영역이 사용자에게 동등하게 중요하지 않음
 - 어떤 사람은 목 디자인이, 어떤 사람은 포켓 사이즈가 중요함
- 기존 방법
 - 하나의 이미지 전체를 고정된 벡터로 변환
 - 상관없는 이미지 영역까지 모델링되어 노이즈가 발생
 - 사람마다 다른 세부적인 선호 특징 요소를 모델링하지 못함
- 본 논문의 제안 방법
 - 분할된 이미지 영역 + 리뷰 데이터 활용
 - 어텐션 메커니즘 활용
 - 사용자가 관심있는 영역 학습

3. Image - VECF



본 논문의 제안 방법

- 분할된 이미지 영역 + 리뷰 데이터 활용
- 어텐션 메커니즘 활용
- 사용자가 관심있는 영역 학습

4. LLM

아마 귀가 닳도록 들었을 것 같습니다.

“LLM”

ChatGPT, GPT-4, PaLM, LLaMA와 같은 모델들이죠

4. LLM

Is ChatGPT a Good Recommender? A Preliminary Study

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출처 : Liu, arxiv chatgpt_rec(2023)

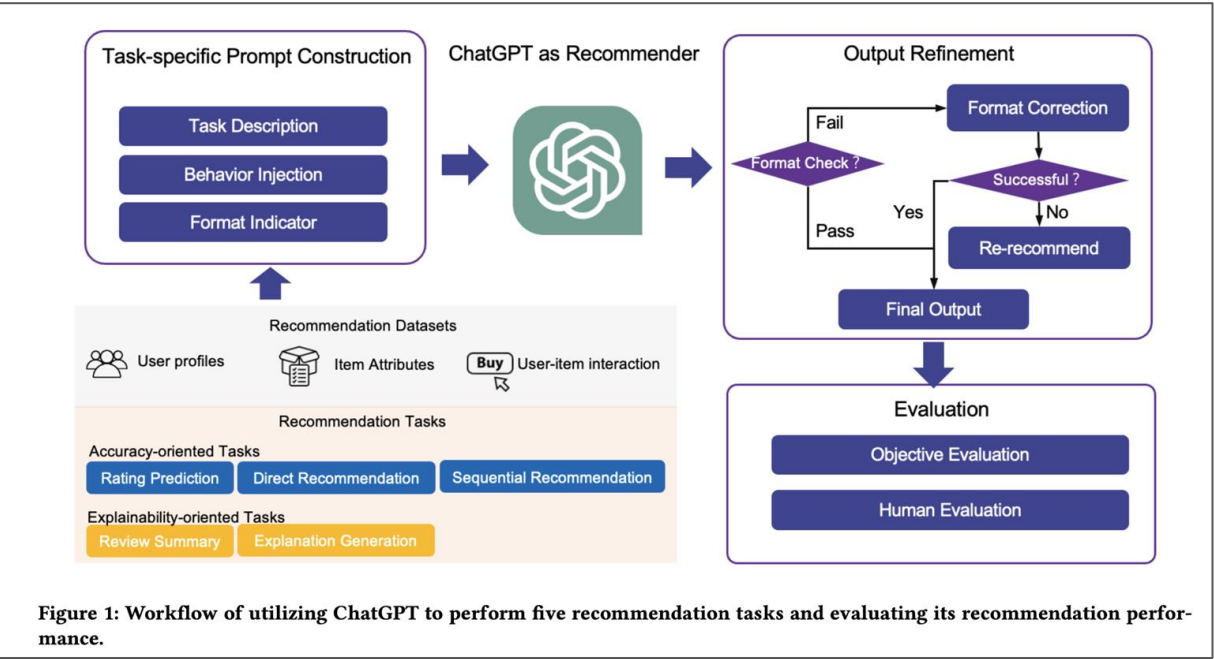


Figure 1: Workflow of utilizing ChatGPT to perform five recommendation tasks and evaluating its recommendation performance.

4. LLM - Chat-REC

Chat-REC: Towards Interactive and Explainable LLMs-Augmented Recommender System

Yunfan Gao¹, Tao Sheng¹, Youlin Xiang¹, Yun Xiong¹, Haofen Wang², and Jiawei Zhang³

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출처 : Gao, arxiv_chatgpt_rec(2023)

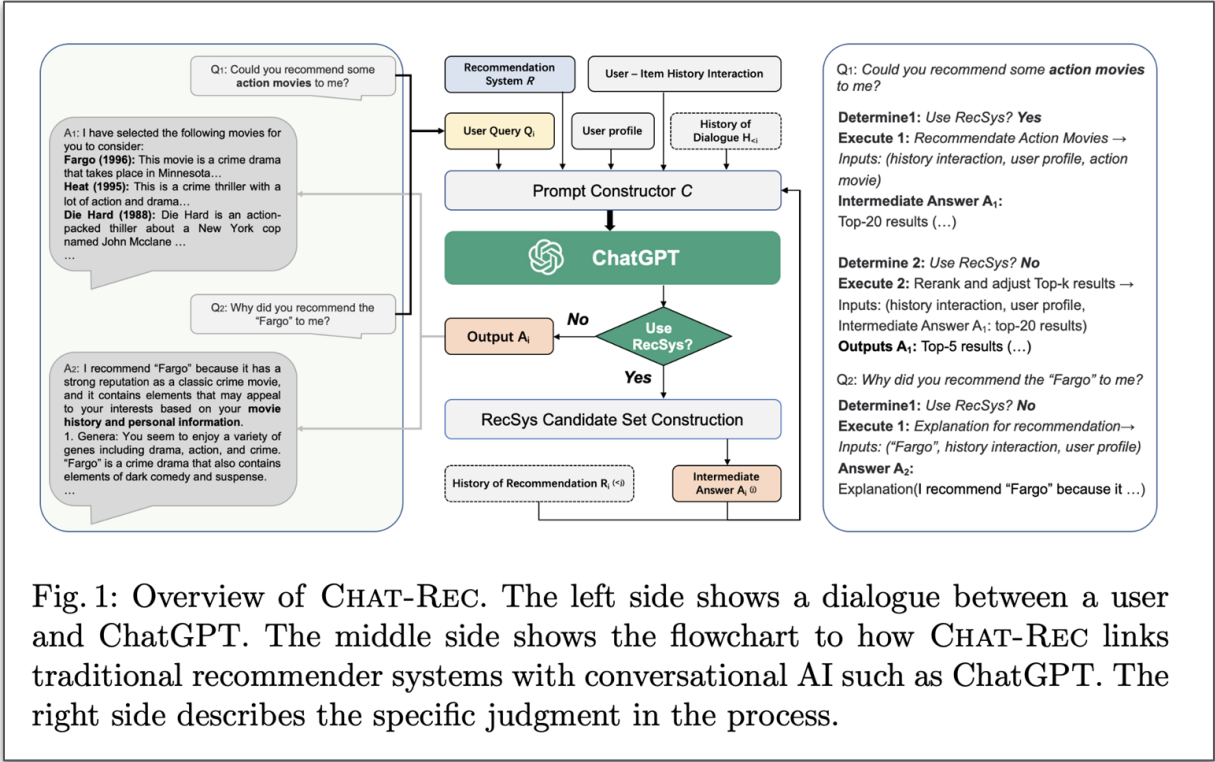


Fig. 1: Overview of CHAT-REC. The left side shows a dialogue between a user and ChatGPT. The middle side shows the flowchart to how CHAT-REC links traditional recommender systems with conversational AI such as ChatGPT. The right side describes the specific judgment in the process.

4. LLM

**LLM과 결합하여 구축하는 추천 시스템은
계속 발전하고 있습니다.**

**LLM과 추천 시스템은 어떻게 발전해 나갈 것인지
재밋게 지켜보면서 우리도 연구해 봅시다.**

핵심 정리

이번 챕터에서는 다음과 같은 내용을 학습하였습니다.

- 협업 필터링의 발전과 NCF 모델
- Shallow 영역과 Deep 영역(FM, DeepFM, Wide & Deep)
- 어텐션과 추천 시스템
- 다양한 딥러닝 기반 추천 시스템(Sequential, Graph, Image, LLM)

다음 시간 소개

다음 시간에는 아래와 같은 내용을 학습할 예정입니다.

- 추천 시스템을 적용하기 위한 작업
- 추천 시스템을 구축하면서 겪을 수 있는 문제들
- 추천 시스템의 관리
- 추천 시스템 엔지니어로서의 커리어

감사합니다.

딥러닝 기반 추천 시스템
다양한 딥러닝 추천 시스템: 특징과 목표

이수진